

Research Association Position Available at University of Pittsburgh: **“High Frequency Transformer Design, Modeling, Prototyping, and Testing”**

Position Opening: Available Immediately

Position: High Frequency Transformer Design, Modeling, Prototyping, and Testing (PhD or MS)

Position Contact: Prof. Paul Ohodnicki, pro8@pitt.edu, Prof. Brandon Grainger, bmg10@pitt.edu

Anticipated Position Start Date: Summer 2026

Position Description:

The Advanced Magnetism for Power and Energy Development Consortium at the University of Pittsburgh invites applications for a research associate with an emphasis on high frequency transformer design, modeling, and testing. Near-term anticipated project work will include modeling and design of medium and high frequency transformers including electromagnetic performance, isolation, thermals, and other design requirements. Additionally, the project will involve prototyping and testing in collaboration with project partners. The position is funded as a result of a recent [ARPA-E award](#) as well as the [Advanced Magnetism for Power and Energy Development](#) consortium based at the University of Pittsburgh. The research associate will have an opportunity to collaborate with an interdisciplinary group of undergraduate, graduate, and PhD level researchers in magnetic materials and electric power conversion applications. In addition, the research associate can expect close collaboration with other researchers from industry and national labs including in collaboration with partners at North Carolina State University, National Laboratory of the Rockies, and Hitachi Energy, Department of Defense, and others.

The advisors will be Prof. Paul Ohodnicki and Prof. Brandon Grainger, and the research associate will have full access to facilities available in the Ohodnicki Lab (<https://www.engineering.pitt.edu/OhodnickiLab/>) at the University of Pittsburgh, the extensive shared facilities of the Advanced Magnetism for Power and Energy Development (AMPED, <https://www.engineering.pitt.edu/AMPED/>) and the GRID Institute ([Home | Energy GRID Institute | University of Pittsburgh](#)), amongst other facilities across campus.

More information about the Ohodnicki Lab research focus and interests can be found here:

<https://www.youtube.com/watch?v=7Vn7XJmHGr4&feature=youtu.be>

Successful applicants should display strong experience in concepts of importance for transformer modeling and design, including concepts and tools such as magnetic equivalent circuits, finite element modeling, applied electromagnetics, and multi-objective optimization methods. Applicants should have a graduate degree (MS or PhD) in electrical engineering, applied physics, or a related field. Additional experience with modeling and design of electrical insulation systems for medium and high voltage systems as well as hands-on characterization and testing of transformers and magnetic components, is desired but not required.

Anticipated project assignments will include establishment of new modeling methods and design approaches to medium and high frequency transformer designs with consideration of electrical insulation and isolation requirements. Implementation of advanced models and integration with multi-objective optimization methods will be a core requirement and participation in transformer prototype testing and characterization for manufactured designs will also be of interest. In addition to conducting research, duties will also include preparing reports, performing literature reviews, supervising and mentoring students, and assisting with other projects in the laboratory. The research associate will also be responsible for overseeing transformer design and characterization efforts that will occur at the [Energy Innovation Center](#).

Application Process:

Interested candidates should contact Prof. Paul Ohodnicki (pro8@pitt.edu) and Prof. Brandon Grainger (bmg10@pitt.edu) with an updated resume at the soonest possible opportunity.